Introduction to Optimal Control

Book: Optimal Control Theory-An Introduction.
Donald E. Kirk, 2004



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Review of Chapters 1&2-Optimal control problem

Find an ***admissible control*** $\boldsymbol{u^*}$ which causes the system

$$\dot{\boldsymbol{x}}(\boldsymbol{t}) = \boldsymbol{a}(\boldsymbol{x}(\boldsymbol{t}), \boldsymbol{u}(\boldsymbol{t}), \boldsymbol{t})$$

to follow an ***admissible trajectory*** $\boldsymbol{x^*}$ that ***minimizes the performance measure***

$$\boldsymbol{J} = \boldsymbol{h}\left(\boldsymbol{x}\left(\boldsymbol{t_f}\right), \boldsymbol{t_f}\right) + \int_{\boldsymbol{t_0}}^{\boldsymbol{t_f}} \mathbf{g}(\boldsymbol{x}(\boldsymbol{t}), \boldsymbol{u}(\boldsymbol{t}), \boldsymbol{t})\boldsymbol{dt}$$

or in Linear Quadratic Regulator (LQR) form

$$\boldsymbol{J} = \int_{\boldsymbol{t_0}}^{\boldsymbol{t_f}} \left[\|\boldsymbol{x}(\boldsymbol{t})\|^2_{\boldsymbol{Q}(\boldsymbol{t})} + \|\boldsymbol{u}(\boldsymbol{t})\|^2_{\boldsymbol{R}(\boldsymbol{t})}\right]\boldsymbol{dt}$$

$\boldsymbol{u^*}$ is called an optimal control and $\boldsymbol{x^*}$ an optimal trajectory.

▪ Two fundamental theories

- **R. E. Bellman (USA)** proposed "**Dynamic programming**" during 1953-1957.

- **L. S. Pontryagin** (**Soviet Union**) developed "**Minimum Principle**" during 1956-1958.

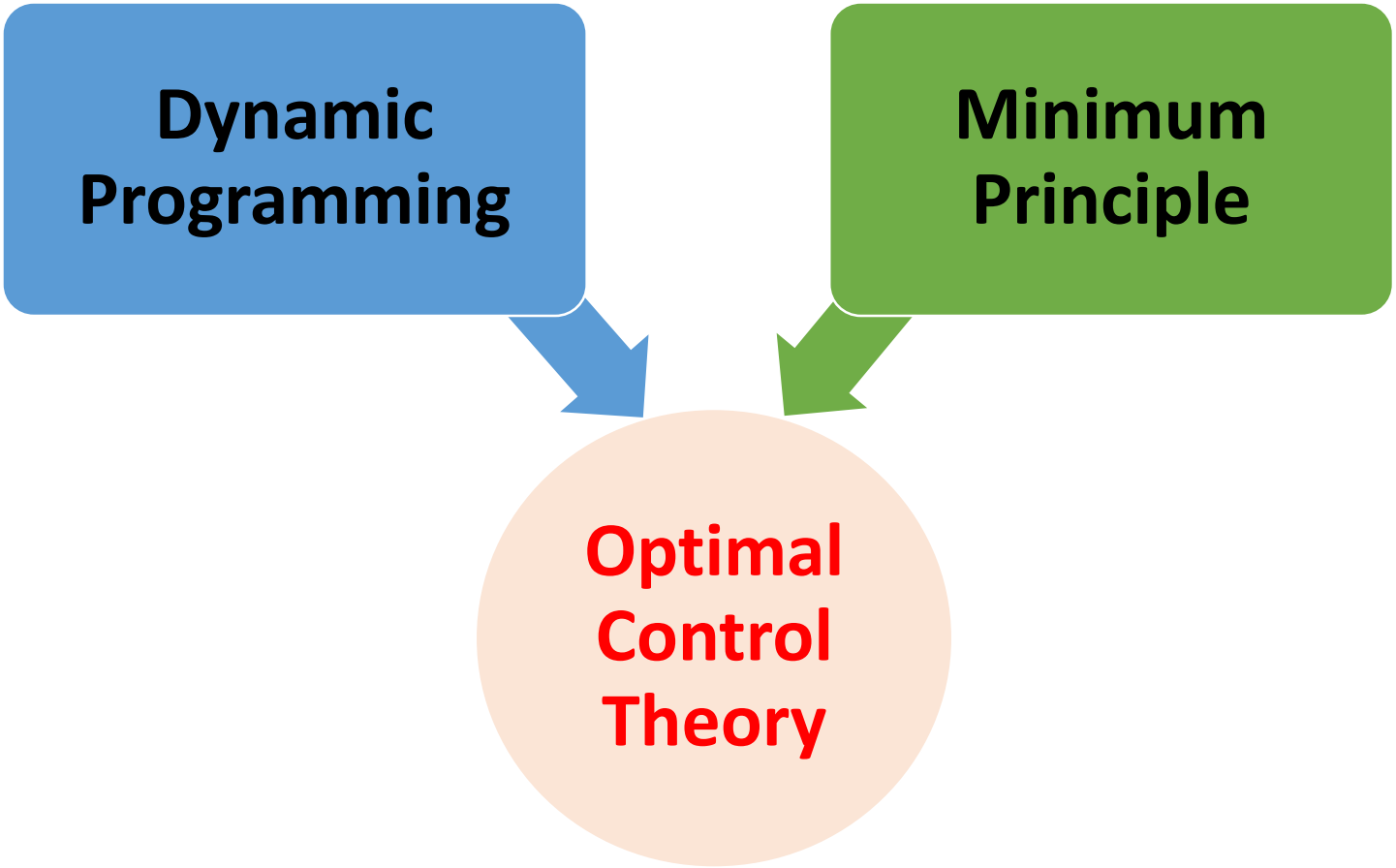


Fig. 1 R. E. Bellman



Fig. 2 L. S. Pontryagin

■ Optimal control theory



Chapter III. Dynamic Programming

1. ***Bellman's Principle of Optimality***
2. **An Optimal Control System**
3. **DP in discrete time**
4. **DP in continuous time**

3.1 Applications-Daily life

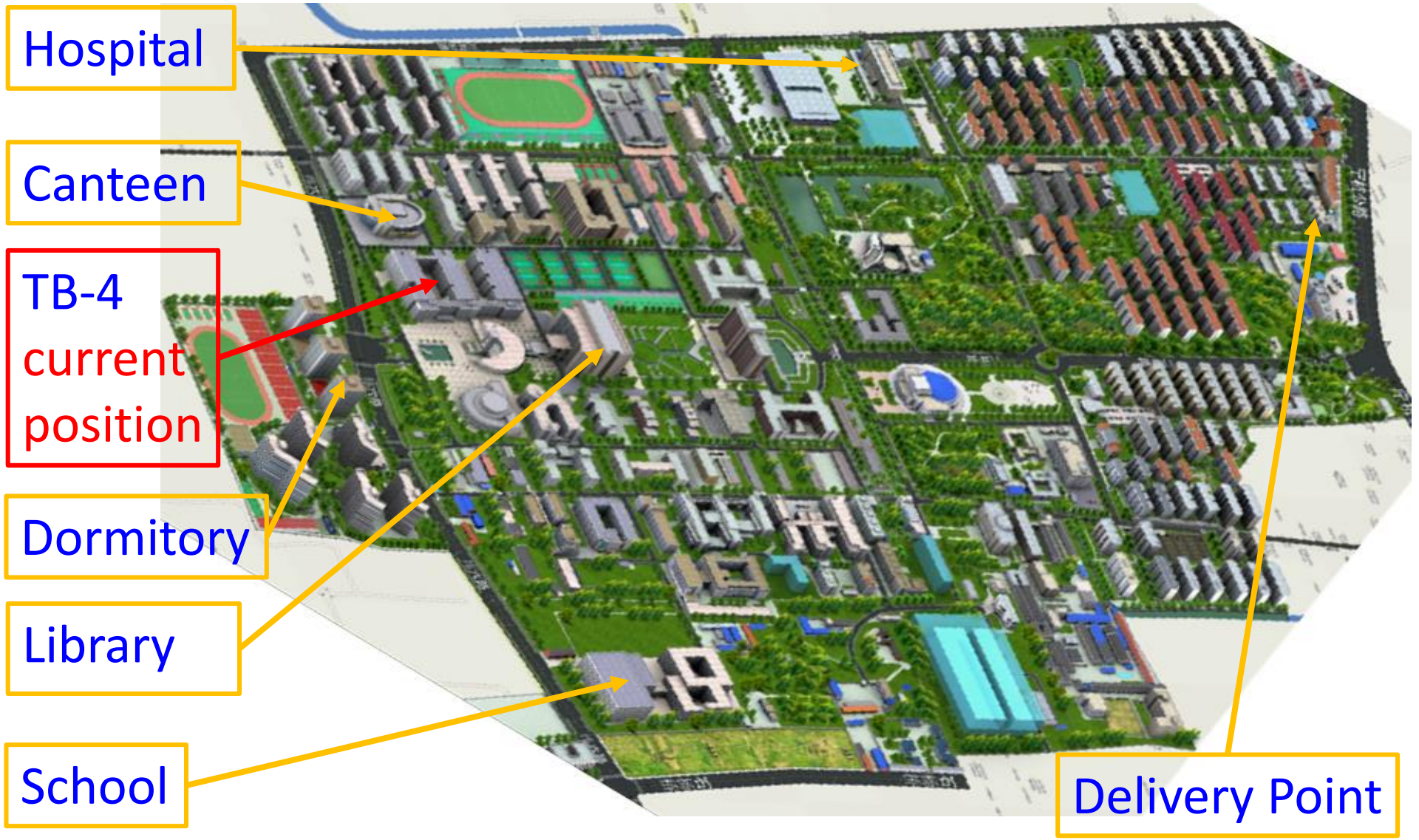


Fig. 3-1 The Map of NJUST

3.2 Principle of Optimality

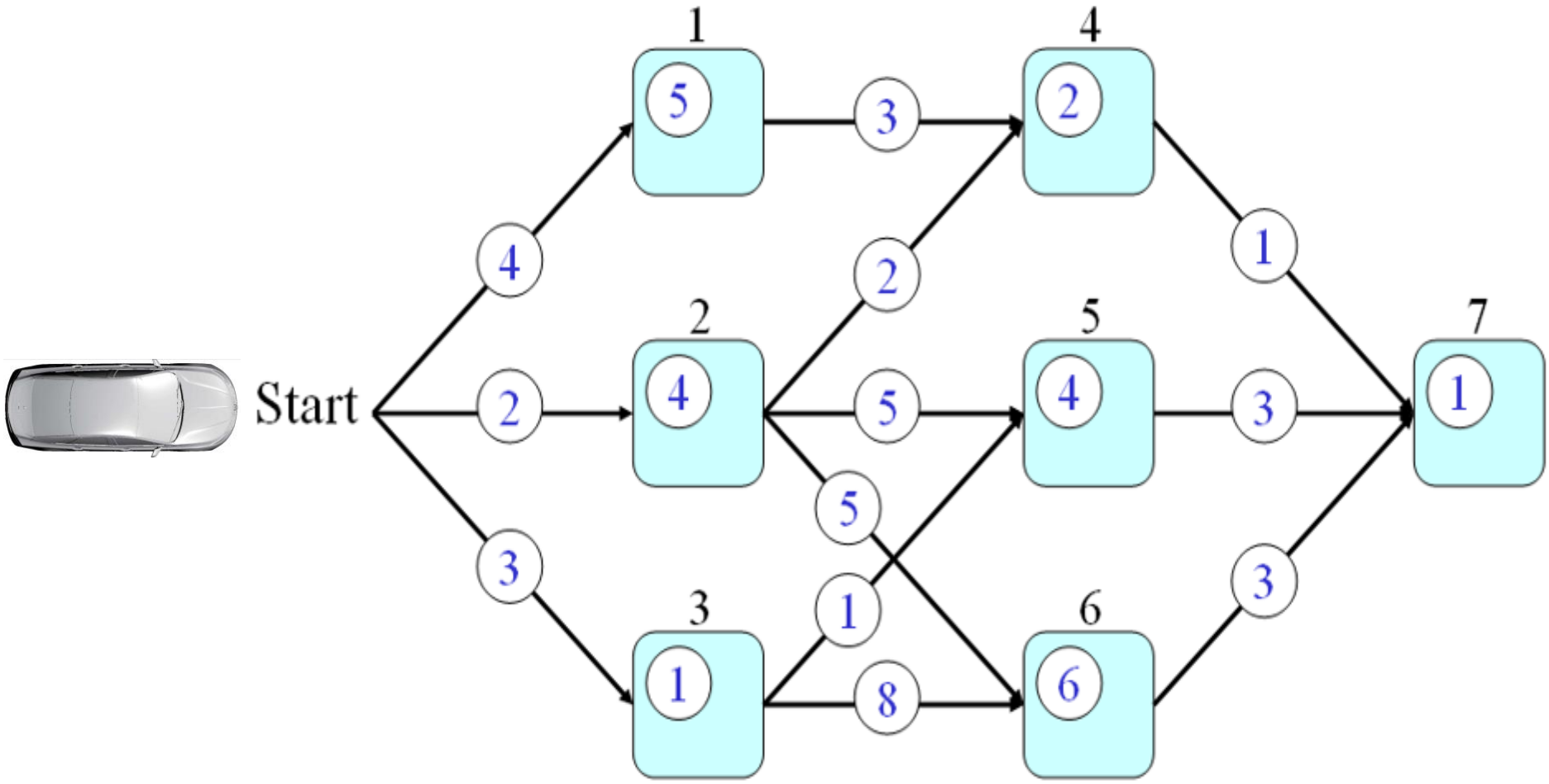
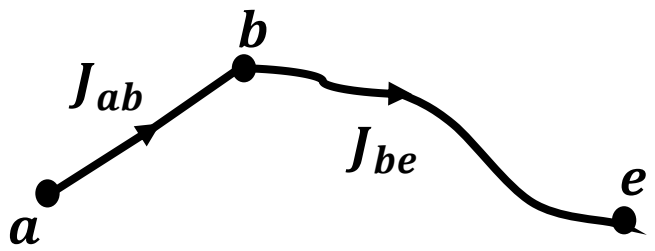


Fig. 3-2 The minimum time from start to intersection 7

3.2 Principle of Optimality

Suppose that the first decision (made at a) results in segment a-b with cost J_{ab} and that the remaining decisions yield segment b-e at a cost of J_{be}. The minimum cost J_{ae}^{*} from a to e is

$$J_{ae}^{*} = J_{ab} + J_{be}.$$



(a)

Fig. 3-3 (a) **Optimal path from a to e**

3.2 Principle of Optimality

Assertion: If $\boldsymbol{a}$-$\boldsymbol{b}$-$\boldsymbol{e}$ is the optimal path from $\boldsymbol{a}$ to $\boldsymbol{e}$, then $\boldsymbol{b}$-$\boldsymbol{e}$ is the optimal path from $\boldsymbol{b}$ to $\boldsymbol{e}$.

Proof ?

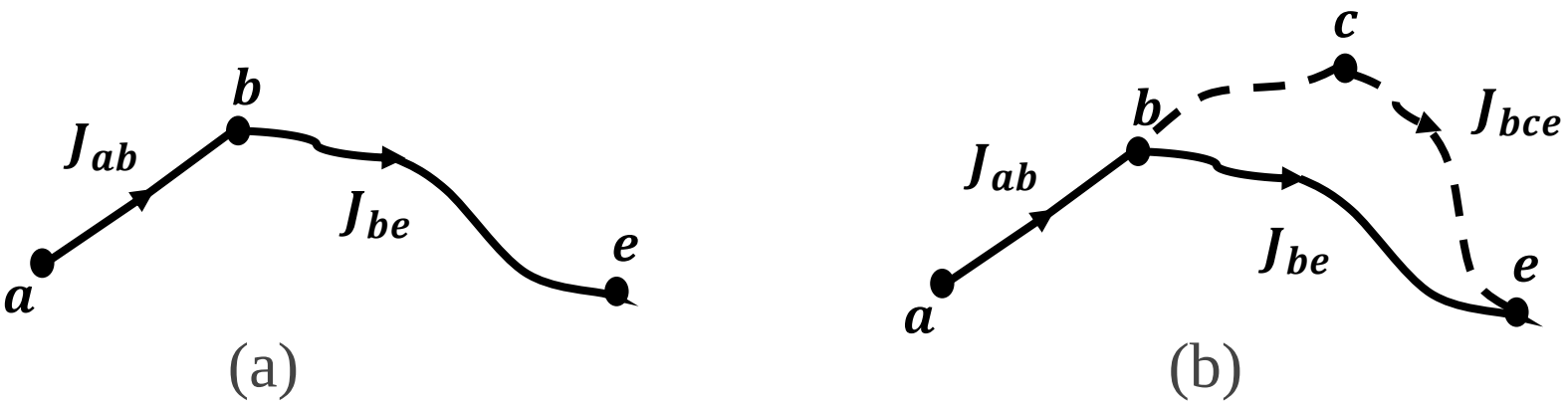


Fig. 3-3 (a) Optimal path from $\boldsymbol{a}$ to $\boldsymbol{e}$. (b) Two possible optimal paths from $\boldsymbol{b}$ to $\boldsymbol{e}$.

3.2 Principle of Optimality

Principle of Optimality

An optimal policy has the property that ***whatever the initial state and initial decision*** are, ***the remaining decisions*** must constitute ***an optimal policy*** with regard to the state resulting from the first decision.

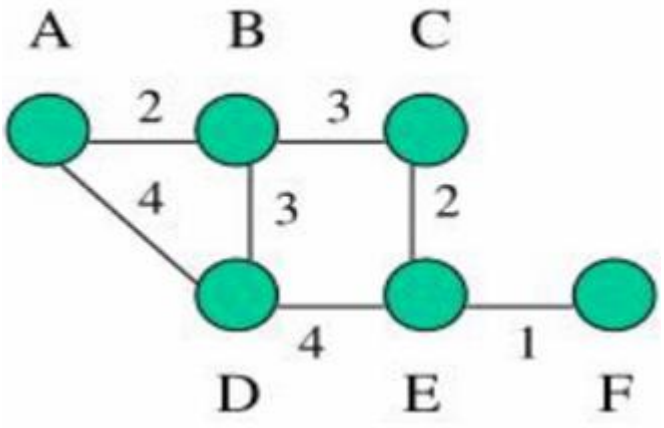


Fig. 3-4 A simple routing graph

1. Optimal path from $\boldsymbol{A}$ to $\boldsymbol{F}$ is $\boldsymbol{A}$-$\boldsymbol{B}$-$\boldsymbol{C}$-$\boldsymbol{E}$-$\boldsymbol{F}$.
2. Then, the optimal path between $\boldsymbol{B}$ and $\boldsymbol{F}$ is $\boldsymbol{B}$-$\boldsymbol{C}$-$\boldsymbol{E}$-$\boldsymbol{F}$.

3.3 Application of the Principle of Optimality

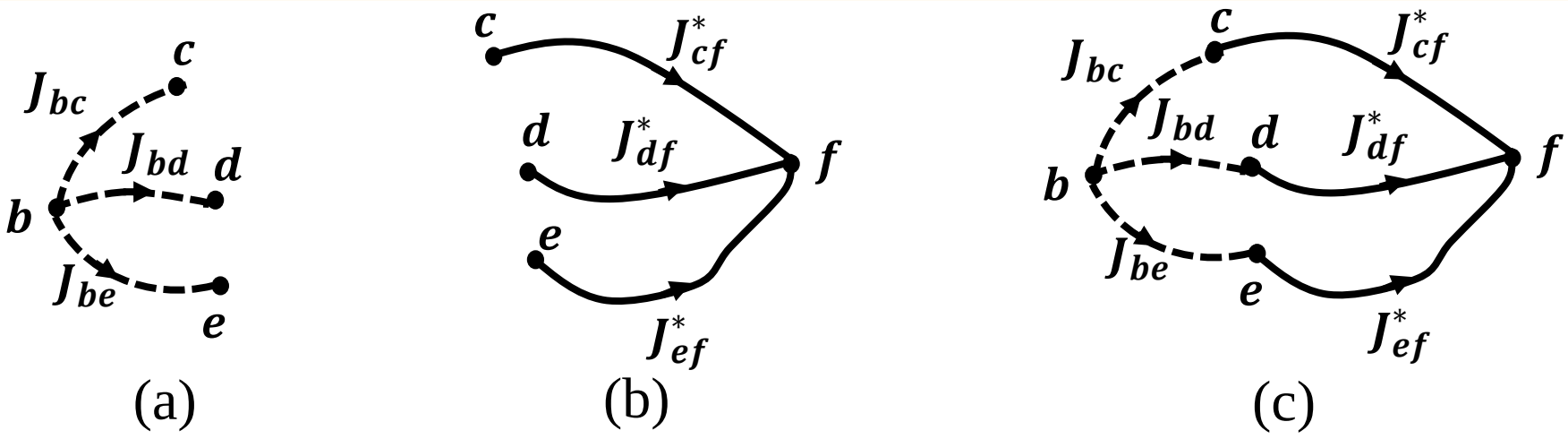


Fig. 3-5 (a) Paths resulting from all allowable decisions at $\boldsymbol{b}$. (b) Optimal paths from $\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}$ to $\boldsymbol{f}$. (c) Candidates for optimal paths from $\boldsymbol{b}$ to $\boldsymbol{f}$

The optimal path that starts at $\boldsymbol{b}$ is found by comparing

$$C^*_{bcf} = J_{bc} + J^*_{cf}$$
$$C^*_{bdf} = J_{bd} + J^*_{df}$$
$$C^*_{bef} = J_{be} + J^*_{ef}$$

$$\Rightarrow J^*_{bf} = \min\{C^*_{bcf}, C^*_{bdf}, C^*_{bef}\}$$

Dynamic **P**rogramming (**DP**) is a computational technique which extends the above decision-making concept to sequences of decisions

3.4 DP applied to a routing problem

A routing problem

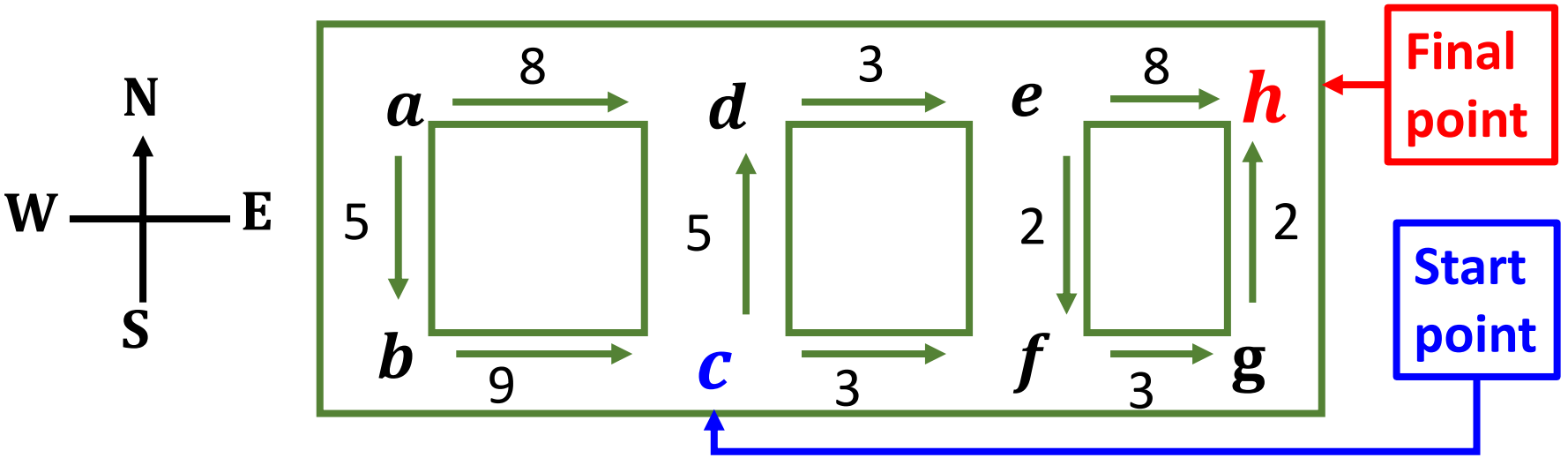


Fig. 3-6 The road map

Problem: Find the shortest path from c to h

1. Exhaustive method,
2. **The method based on Principle of Optimality?**

3.4 DP applied to a routing problem

A routing problem

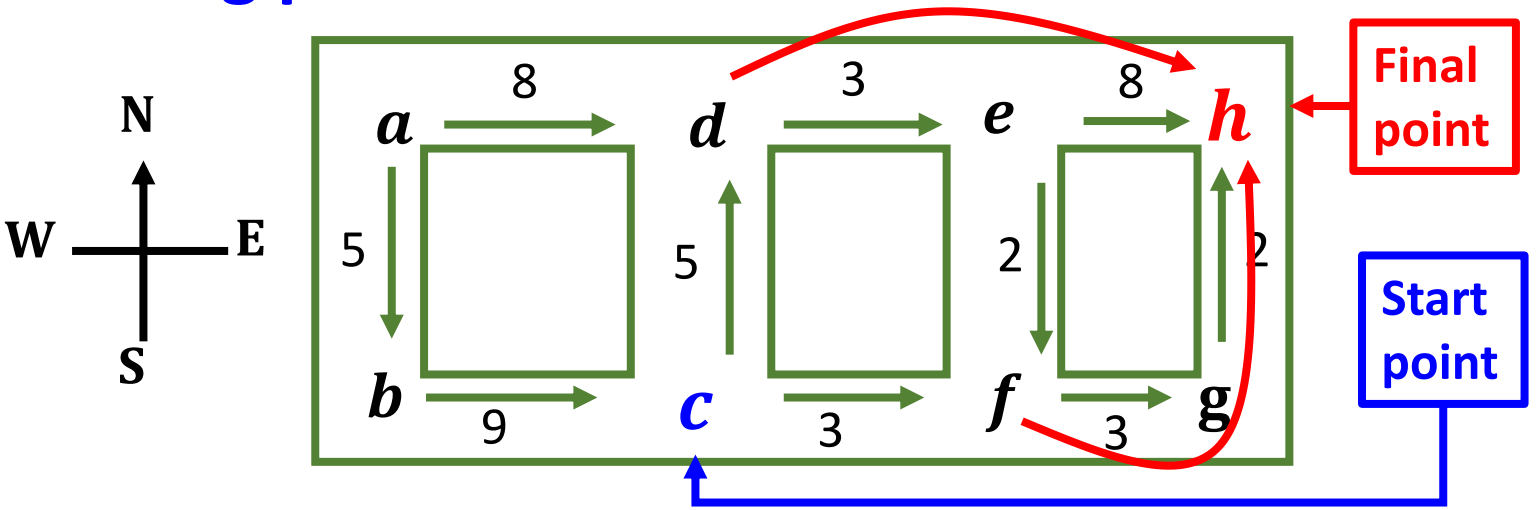


Fig. 3-6 The road map

Step 1： J^*_{dh} denotes the shortest path from d to h.Suppose J^*_{dh} and J^*_{fh} are already known, then

$$C^*_{cdh} = J_{cd} + J^*_{dh},$$
$$C^*_{cfh} = J_{cf} + J^*_{fh},$$
$$J^*_{ch} = min\{C^*_{cdh}, C^*_{cfh}\}.$$

3.4 DP applied to a routing problem

A routing problem

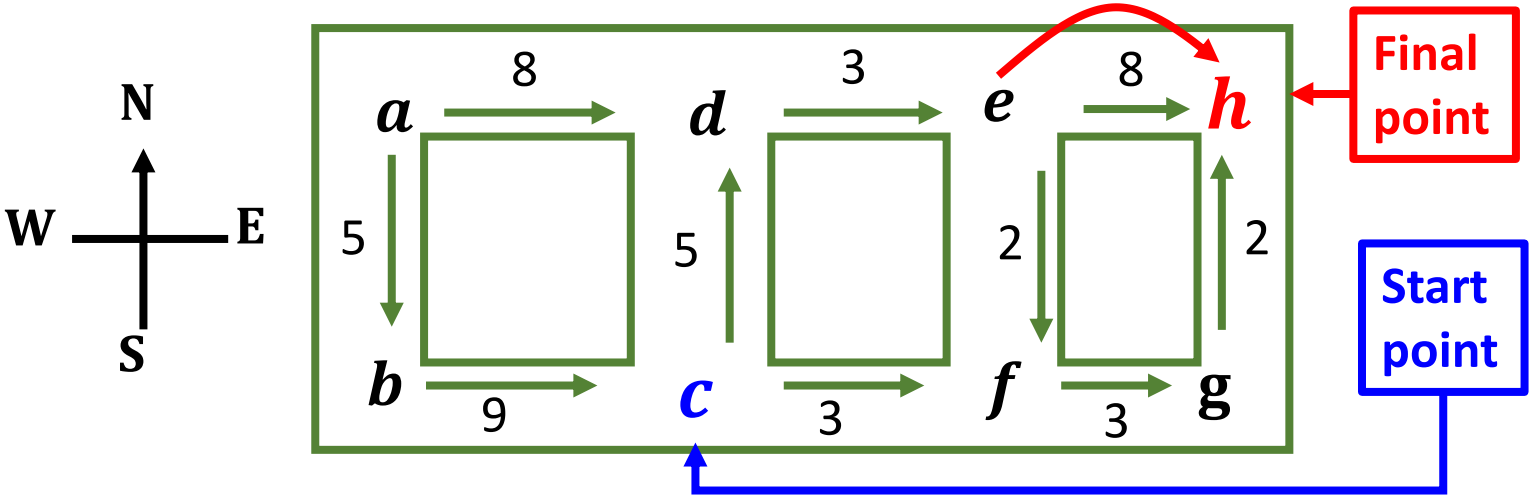


Fig. 3-6 The road map

Step 2：To calculate J^*_{dh}, we have to know J^*_{eh}

$$J^*_{dh} = J_{de} + J^*_{eh}.$$

Step 3：Calculate J^*_{eh}

$$J^*_{eh} = min\{J_{eh},\ J_{efgh}\}.$$

3.4 DP applied to a routing problem

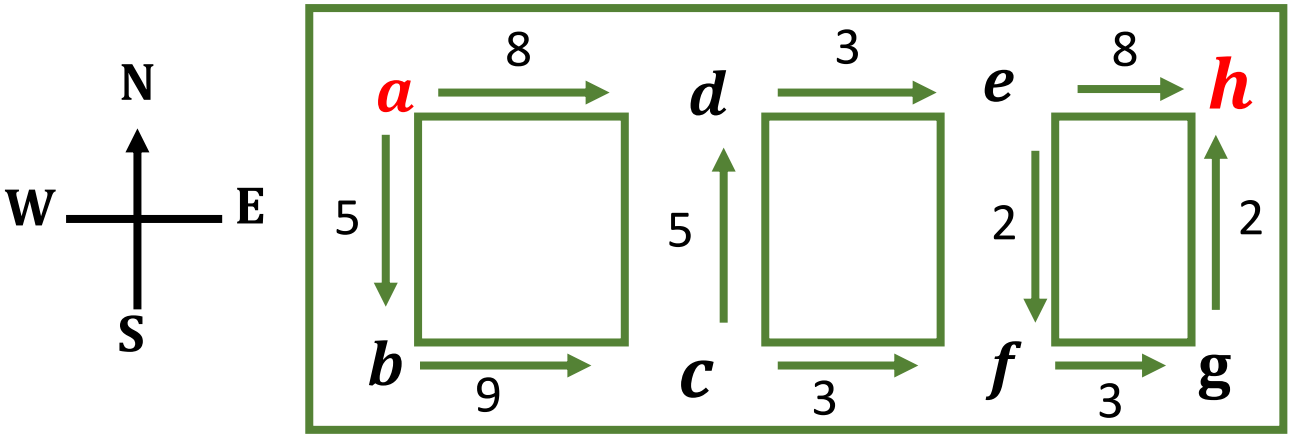


Fig. 3-6 The road map

Question: The shortest path from a to h?

3.4 DP applied to a routing problem

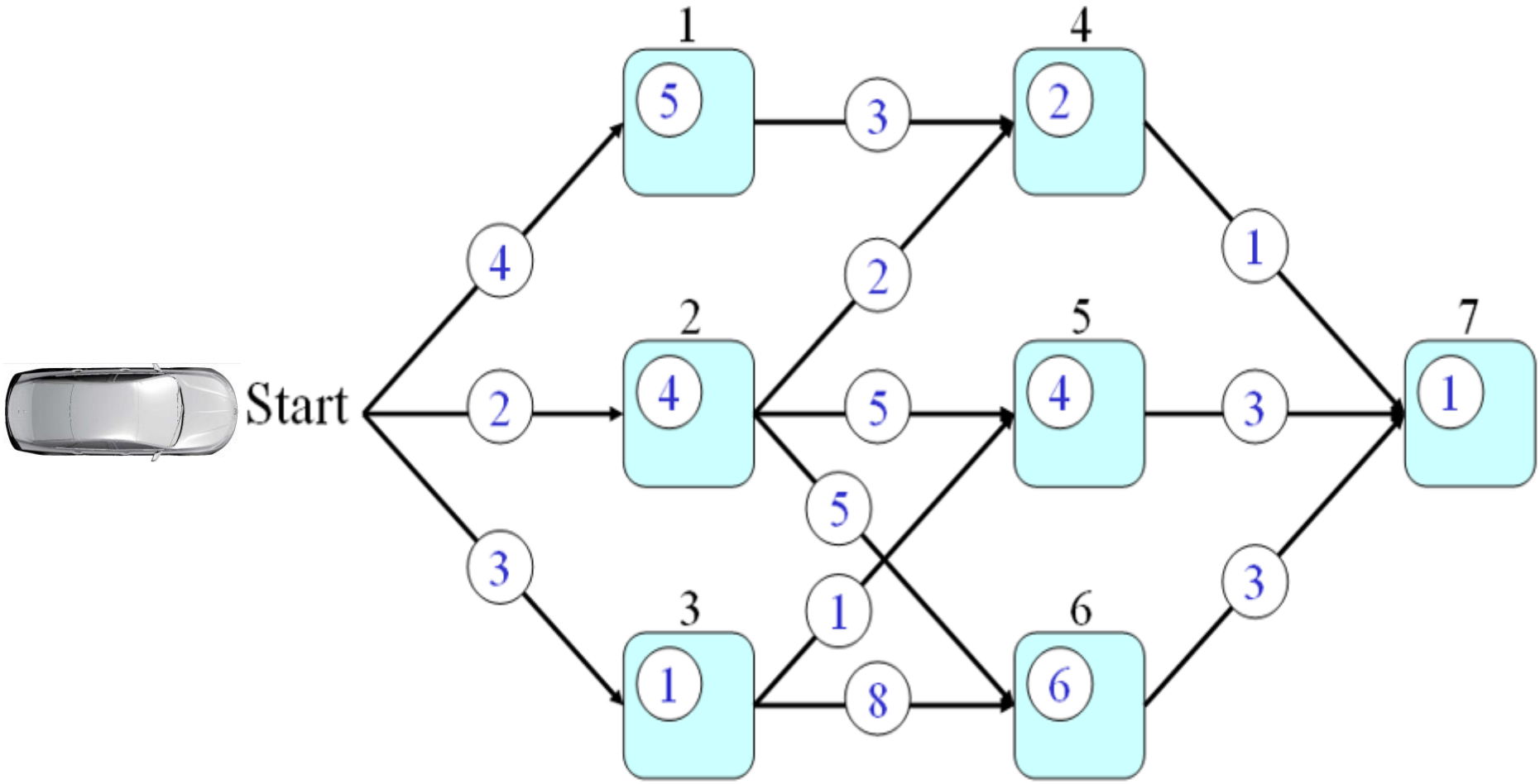


Fig. 3-7 The minimum time from start to intersection 7

3.4 DP applied to a routing problem

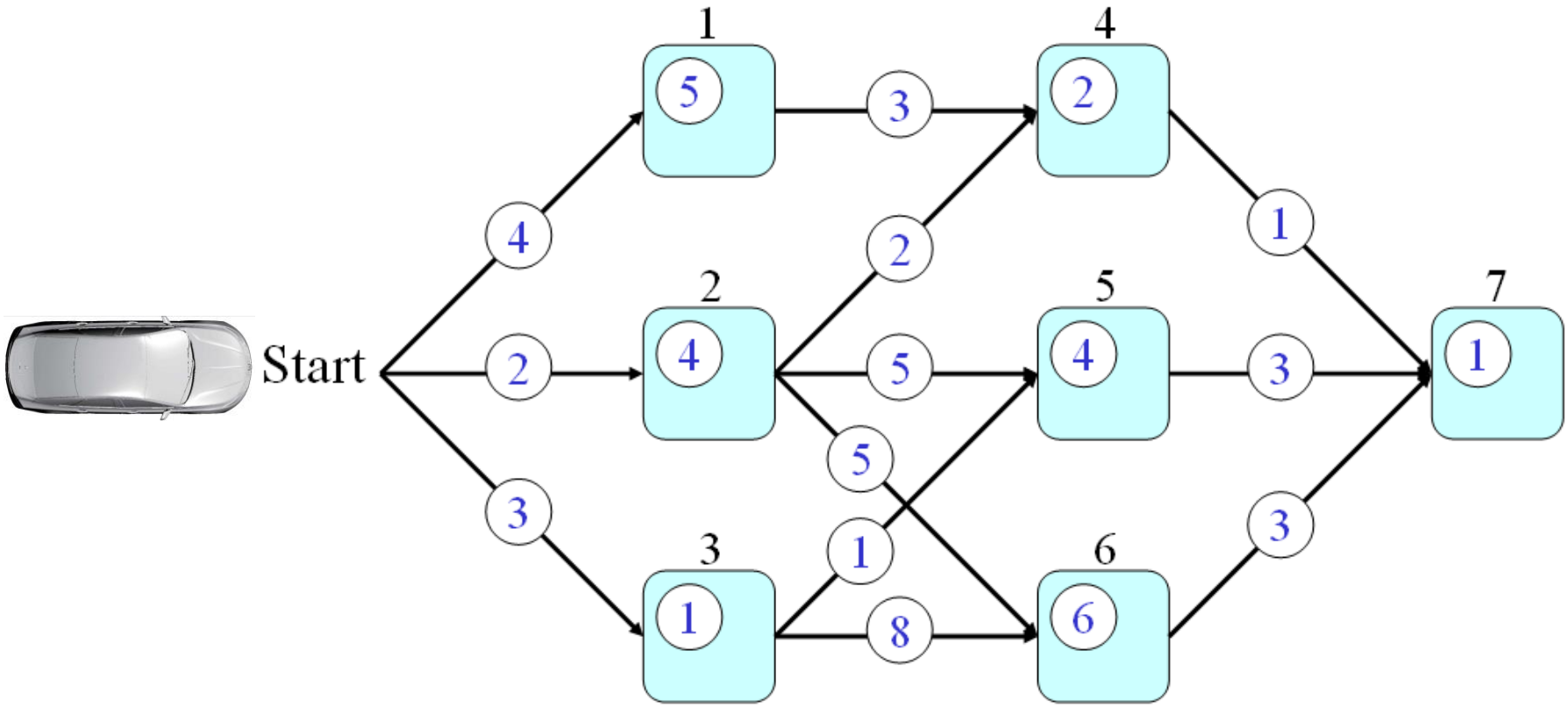


Fig. 3-7 The minimum time from start to intersection 7

3.4 DP applied to a routing problem

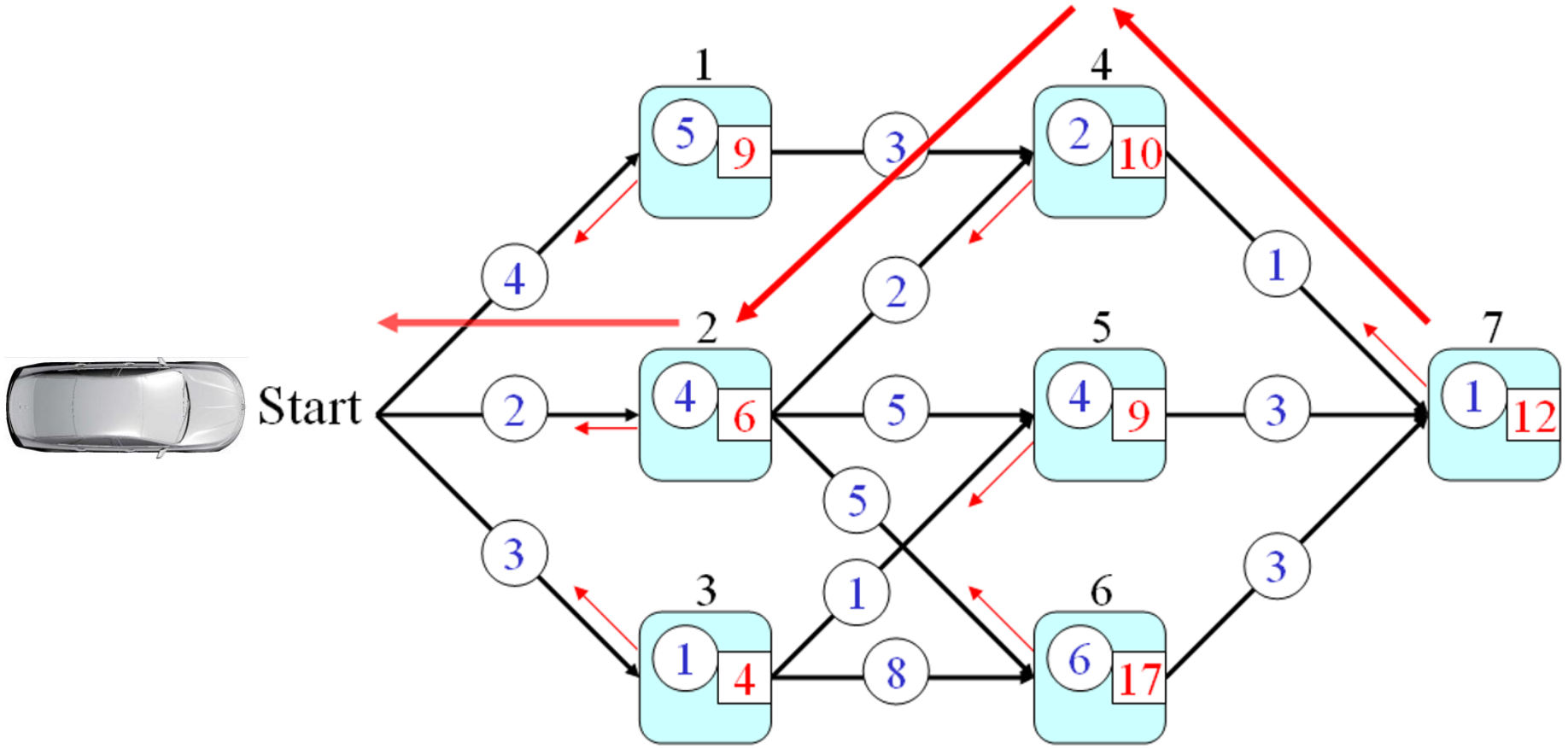


Fig. 3-7 The minimum time from start to intersection 7

3.4 DP applied to a routing problem

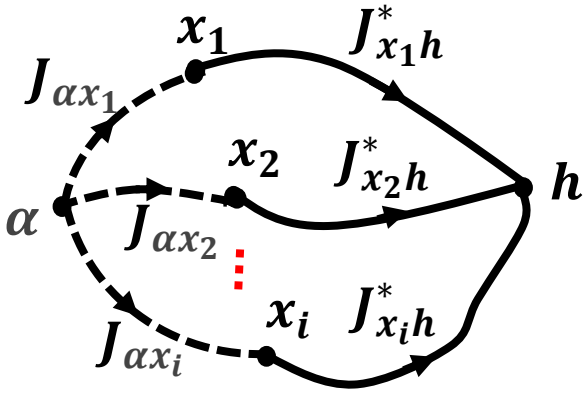


Fig. 3-8 Candidates for optimal paths from α to h.

The principle of optimality implies

$$C^*_{\alpha x_i h} = J_{\alpha x_i} + J^*_{x_i h} .$$

The optimal decision at α is $u^*(\alpha)$, which leads to

$$J^*_{\alpha h} = min\{C^*_{\alpha x_1 h}, C^*_{\alpha x_2 h}, \cdots, C^*_{\alpha x_i h}, \cdots\}.$$

The above two equations define the algorithm called **Dynamic Programming (DP)**.

3.4 DP applied to a routing problem

Exhaustive method *VS* Dynamic Programming （DP）

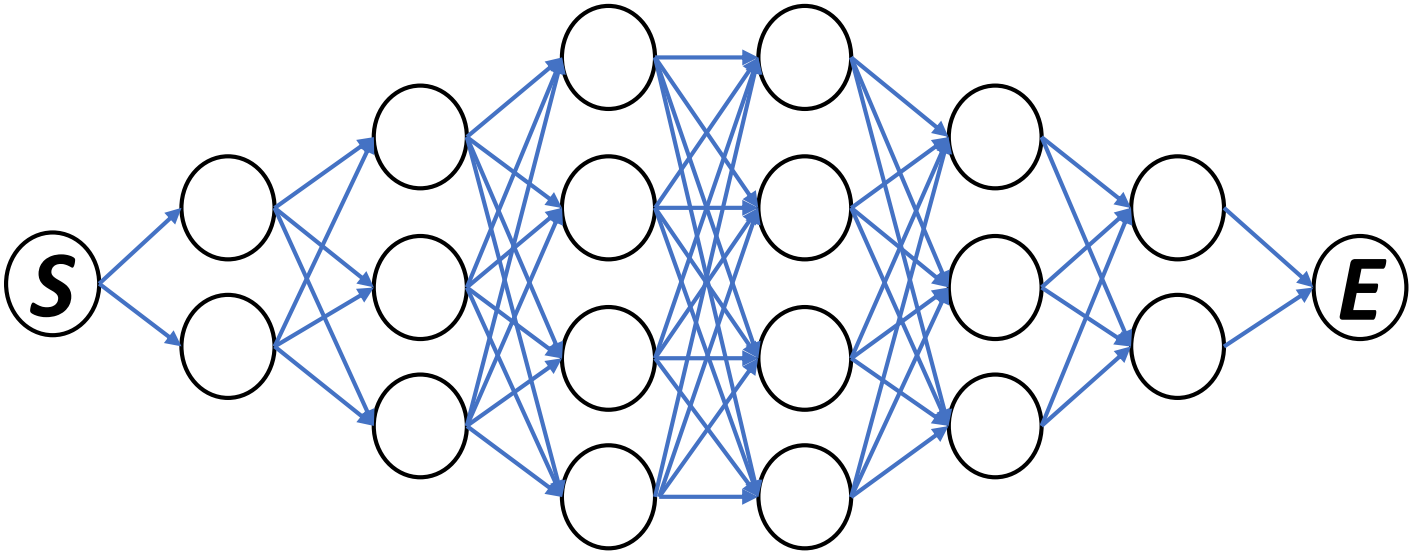


Fig. 3-9 A routing map

1. Exhaustive method: 1*2*3*4*4*3*2*1= 576

2. DP: 2*3 + 3*4+ 4*4+ 4*3 + 3*2+ 2*1 = 54

3.4 DP applied to a routing problem

Application



Fig. 3-10 Intelligent scheduling system

3.4 DP applied to a routing problem

Remarks:

- ***DP*** is proposed based on the ***principle of optimality***,
- ***DP*** can ***reduce the computational complexity***.

3.5 An optimal control system

Consider a system described by the first-order D.E.

$$\dot{x}\,(t) = ax(t) + bu(t).$$

The admissible values of the control and state are constrained by

$$0.0 \le x(t) \le 1.5,$$
$$-1.0 \le u(t) \le 1.0,$$

and the performance measure to be minimized is

$$J = x^2(T) + \lambda \int_0^T u^2(t)dt,$$

where T is the final time and λ is a weighting factor.

3.5 An optimal control system

Discretization

The difference equation is

$$x(k+1) = (1 + a\Delta t)x(k) + b\Delta t u(k)$$

and the performance measure to be minimized is

$$J = x^2(N) + \lambda\Delta t \sum_{k=0}^{N-1} u^2(k)\,.$$

Let $a = 0, b = 1, \lambda = 2, T = 2, \Delta t = 1, N = 2.$

3.5 An optimal control system

The difference equation is

$$x(k+1) = x(k) + u(k).$$

The admissible values of the control and state are constrained by

$$0.0 \le x(k) \le 1.5, \qquad k = 0, 1, 2,$$

$$-1.0 \le u(k) \le 1.0, \qquad k = 0, 1,$$

and the performance measure to be minimized is

$$J = x^2(2) + 2u^2(0) + 2u^2(1).$$

Note that the state $x(k)$ is quantized to be $(\mathbf{0}, \mathbf{0.5}, \mathbf{1.0}, \mathbf{1.5})$ and the control $u(k)$ is quantized to $(\mathbf{-1.0}, \mathbf{-0.5}, \mathbf{0}, \mathbf{0.5}, \mathbf{1.0})$.

3.5 An optimal control system

The difference equation is

$$x(k+1) = x(k) + u(k).$$

The admissible values of the control and state are constrainedand the performance measure to be minimized is

$$J = x^2(2) + 2u^2(0) + 2u^2(1).$$

The state $x(k)$ is quantized to be $(\mathbf{0}, \mathbf{0.5}, \mathbf{1.0}, \mathbf{1.5})$ and the control $u(k)$ is quantized to $(\mathbf{-1.0}, \mathbf{-0.5}, \mathbf{0}, \mathbf{0.5}, \mathbf{1.0})$.

Step 1: Put $k = 1$, select **one of the quantized values** of $x(1)$, try all the quantized values of $u(1)$

Step 2: Repeat the procedure in Step 1 for the **remaining quantized values** of $x(1)$

Table 3-2 COSTS OF OPERATION OVER THE LAST STAGE

Current state $x(1)$	*Control* $u(1)$	*Next state* $x(2) = x(1) + u(1)$	*Cost* $x^2(2) + 2u^2(1) = J_{12}(x(1), u(1))$	*Minimum cost* $J_{12}^*(x(1))$	*Optimal control applied at* $k = 1$ $u^*(x(1), 1)$
1.5	0.0	1.5	$(1.5)^2 + 2(0.0)^2 = 2.25$		
	−0.5	1.0	$(1.0)^2 + 2(-0.5)^2 = 1.50$	$J_{12}^*(1.5) = 1.50$	$u^*(1.5, 1) = -0.5$
	−1.0	0.5	$(0.5)^2 + 2(-1.0)^2 = 2.25$		
1.0	0.5	1.5	$(1.5)^2 + 2(0.5)^2 = 2.75$		
	0.0	1.0	$(1.0)^2 + 2(0.0)^2 = 1.00$		
	−0.5	0.5	$(0.5)^2 + 2(-0.5)^2 = 0.75$	$J_{12}^*(1.0) = 0.75$	$u^*(1.0, 1) = -0.5$
	−1.0	0.0	$(0.0)^2 + 2(-1.0)^2 = 2.00$		
0.5	1.0	1.5	$(1.5)^2 + 2(1.0)^2 = 4.25$		
	0.5	1.0	$(1.0)^2 + 2(0.5)^2 = 1.50$		
	0.0	0.5	$(0.5)^2 + 2(0.0)^2 = 0.25$	$J_{12}^*(0.5) = 0.25$	$u^*(0.5, 1) = 0.0$
	−0.5	0.0	$(0.0)^2 + 2(-0.5)^2 = 0.50$		
0.0	1.0	1.0	$(1.0)^2 + 2(1.0)^2 = 3.00$		
	0.5	0.5	$(0.5)^2 + 2(0.5)^2 = 0.75$		
	0.0	0.0	$(0.0)^2 + 2(0.0)^2 = 0.00$	$J_{12}^*(0.0) = 0.00$	$u^*(0.0, 1) = 0.0$

Table 3-3 COSTS OF OPERATION OVER THE LAST TWO STAGES

Current state $x(0)$	*Control* $u(0)$	*Next state* $x(1) = x(0) + u(0)$	*Minimum cost over last two stages for trial value* $u(0)$ $J_{01}(x(0), u(0)) + J_{12}^*(x(1)) = 2u^2(0) + J_{12}^*(x(1)) = C_{02}^*(x(0), u(0))$	*Minimum cost over last two stages* $J_{02}^*(x(0))$	*Optimal control applied at* $k = 0$ $u^*(x(0), 0)$
1.5	0.0	1.5	$2(0.0)^2 + 1.50 = 1.50$		
	−0.5	1.0	$2(-0.5)^2 + 0.75 = 1.25$	$J_{02}^*(1.5) = 1.25$	$u^*(1.5, 0) = -0.5$
	−1.0	0.5	$2(-1.0)^2 + 0.25 = 2.25$		
1.0	0.5	1.5	$2(0.5)^2 + 1.50 = 2.00$		
	0.0	1.0	$2(0.0)^2 + 0.75 = 0.75$	$J_{02}^*(1.0) = \begin{Bmatrix} 0.75 \\ 0.75 \end{Bmatrix}$	$u^*(1.0, 0) = \begin{Bmatrix} 0.0 \\ -0.5 \end{Bmatrix}$
	−0.5	0.5	$2(-0.5)^2 + 0.25 = 0.75$		
	−1.0	0.0	$2(-1.0)^2 + 0.00 = 2.00$		
0.5	1.0	1.5	$2(1.0)^2 + 1.50 = 3.50$		
	0.5	1.0	$2(0.5)^2 + 0.75 = 1.25$		
	0.0	0.5	$2(0.0)^2 + 0.25 = 0.25$	$J_{02}^*(0.5) = 0.25$	$u^*(0.5, 0) = 0.0$
	−0.5	0.0	$2(-0.5)^2 + 0.00 = 0.50$		
0.0	1.0	1.0	$2(1.0)^2 + 0.75 = 2.75$		
	0.5	0.5	$2(0.5)^2 + 0.25 = 0.75$		
	0.0	0.0	$2(0.0)^2 + 0.00 = 0.00$	$J_{02}^*(0.0) = 0.00$	$u^*(0.0, 0) = 0.0$

3.5 An optimal control system

The summary of the example

$$C^*_{02}(x(0),u(0)) = J_{01}(x(0),u(0)) + J^*_{12}(x(1))$$
$$J^*_{02}(x(0)) = \min_{u(0)}[J_{01}(x(0),u(0)) + J^*_{12}(x(1))]$$

The functional equation of dynamic programming

$$C^*_{kN}(x(k),u(k)) = J_{k,k+1}(x(k),u(k)) + J^*_{k+1,N}(x(k+1)),$$
$$J^*_{kN}(x(k)) = \min_{u(k)}[C^*_{kN}(x(k),u(k))].$$

3.6 Interpolation

The difference equation is

$$x(k+1) = x(k) + u(k).$$

The the admissible values of the control and state are constrained by

$$0.0 \le x(k) \le 1.5, \qquad k = 0, 1, 2,$$

$$-1.0 \le u(k) \le 1.0, \qquad k = 0, 1, 2,$$

and the performance measure to be minimized is

$$J = x^2(2) + 2u^2(0) + 2u^2(1).$$

Question: The state $x(k)$ is quantized to be $(\mathbf{0}, \mathbf{0.5}, \mathbf{1.0}, \mathbf{1.5})$ and the control $u(k)$ is quantized to $(\mathbf{-1.0}, \mathbf{-0.75} - \mathbf{0.5}, \mathbf{-0.25}, \mathbf{0}, \mathbf{0.25}, \mathbf{0.5}, \mathbf{0.75}, \mathbf{1.0})$.

The initial state $x(0) = 1.5$

3.6 Interpolation

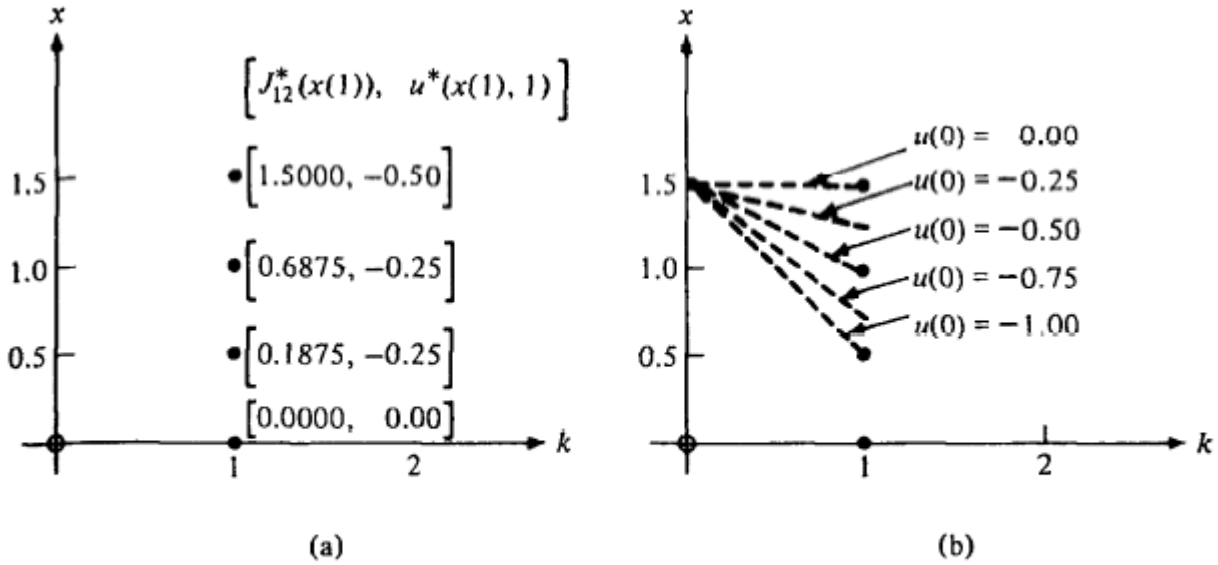


Figure 3-4 (a) Minimum costs and optimal controls for quantized values of $x(1)$. (b) Paths resulting from the application of quantized control values at $x(0) = 1.5$

By **Linear interpolation**, one has

$$J_{12}^*(1.25) = 0.6875 + 0.5(1.5 - 0.6875) = 1.09375.$$

$$J_{12}^*(0.75) = 0.1875 + 0.5(0.6875 - 0.1875) = 0.43750.$$

3.6 Interpolation

Table 3-4 COSTS OF OPERATION OVER THE LAST TWO STAGES FOR $x(0) = 1.50$

Current state $x(0)$	*Control* $u(0)$	*Next state* $x(1) = x(0) + u(0)$	*Minimum cost over last two stages for trial value* $u(0)$: $J_{01}(x(0), u(0)) + J_{12}^*(x(1)) =$ $2u^2(0) + J_{12}^*(x(1)) = C_{02}^*(x(0), u(0))$	*Minimum cost over last two stages* $J_{02}^*(x(0))$	*Optimal control applied at* $k = 0$ $u^*(x(0), 0)$
1.50	0.00	1.50	$2(0.00)^2 + 1.50000 = 1.50000$		
	−0.25	1.25	$2(-0.25)^2 + 1.09375 = 1.21875$		
	−0.50	1.00	$2(-0.50)^2 + 0.68750 = 1.18750$	$J_{02}^*(1.5) = 1.18750$	$u^*(1.5, 0) = -0.50$
	−0.75	0.75	$2(-0.75)^2 + 0.43750 = 1.56250$		
	−1.00	0.50	$2(-1.00)^2 + 0.18750 = 2.18750$		

$J_{02}^*(x(0) = 1.5) = 1.18750$ *with optimal control* $u^*(0) = -0.5$, $u^*(1) = -0.25$.